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quantum-inspired algorithms

quantum algorithms

machine learning

finance

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TITLE	CITED BY	YEAR
Experimental demonstration of quantum advantage in communication complexity for Euclidean distance problem V Yacoub, N Kumar, I Kerenidis, E Diamanti arXiv preprint arXiv:2605.31516		2026
Entropy Distribution as a Fingerprint for Hallucinations in Generative Models MJ Villani, P Deshpande, A Seshadri, R Yalovetzky, N Kumar arXiv preprint arXiv:2605.28264		2026
Anytime Training with Schedule-Free Spectral Optimization A Apte, P Deshpande, N Kumar, S Chakrabarti, JL Kim arXiv preprint arXiv:2605.23061		2026
A unified framework for provably efficient algorithms to estimate shapley values T Chen, A Seshadri, MJ Villani, P Niroula, S Chakrabarti, A Ray, ... Advances in Neural Information Processing Systems 38, 164943-164995	3	2026
Systems and methods for providing secure differential privacy using certifiable randomness protocols A Lang, M Pistoia, S Chakrabarti, N Kumar, P NIROULA, R Shaydulin US Patent 12,524,555		2026
Systems and methods for generating jointly certifiable randomness and providing jointly certifiable randomness via publicly-certifiable randomness beacons A Omar, S Chakrabarti, N Kumar, P NIROULA, R Shaydulin, M Pistoia, ... US Patent App. 18/680,190		2025
Realization of a Quantum Streaming Algorithm on Long-lived Trapped-ion Qubits P Niroula, S Chakrabarti, S Kordonowy, N Kumar, S Omanakuttan, ... arXiv preprint arXiv:2511.03689	2	2025
Applications of certified randomness O Amer, S Chakrabarti, K Chakraborty, S Eloul, N Kumar, C Lim, M Liu, ... Nature Reviews Physics 7 (9), 514-524	6	2025
A simple analysis of a quantum-inspired algorithm for solving low-rank linear systems T Chen, JL Kim, A Ray, S Chakrabarti, D Herman, N Kumar		2025

TITLE	CITED BY	YEAR
arXiv preprint arXiv:2508.13108		
Systems and methods for construction and retraining of decision trees ensemble using hybrid classical-quantum algorithms R YALOVETZKY, N Kumar, P MINSEN, C Li, M Pistoia US Patent App. 18/427,223		2025
Systems and methods for approximation of shapley values in memory-constrained environments P NIROULA, N Kumar, S Chakrabarti, R YALOVETZKY, M Pistoia US Patent App. 18/427,467		2025
MetaTT: A global tensor-train Adapter for parameter-efficient fine-tuning J Lopez-Piqueres, P Deshpande, A Ray, MJ Villani, M Pistoia, N Kumar arXiv preprint arXiv:2506.09105	6	2025
GPU-Parallelizable Randomized Sketch-and-Precondition for Linear Regression using Sparse Sign Sketches T Chen, P Niroula, A Ray, P Subrahmanya, M Pistoia, N Kumar arXiv preprint arXiv:2506.03070	4	2025
Characterizing privacy in quantum machine learning J Heredge, N Kumar, D Herman, S Chakrabarti, R Yalovetzky, ... npj Quantum Information 11 (1), 80	6	2025
Provably faster randomized and quantum algorithms for k-means clustering via uniform sampling T Chen, A Ray, A Seshadri, D Herman, B Bach, P Deshpande, A Som, ... arXiv preprint arXiv:2504.20982	3	2025
A numerical gradient inversion attack in variational quantum neural-networks G Papadopoulos, S Eloul, Y Satsangi, J Heredge, N Kumar, CF Chen, ... arXiv preprint arXiv:2504.12806	3	2025
Certified randomness using a trapped-ion quantum processor M Liu, R Shaydulin, P Niroula, M DeCross, SH Hung, WY Kon, ... Nature 640 (8058), 343-348	82	2025
New Improvements in Solving Large LABS Instances Using Massively Parallelizable Memetic Tabu Search Z Zhang, J Shen, N Kumar, M Pistoia arXiv preprint arXiv:2504.00987	1	2025
Systems and methods for decision tree construction and update using quantum algorithms R YALOVETZKY, N Kumar, C Li, M Pistoia US Patent App. 18/448,662		2025
Des-q: a quantum algorithm to provably speedup retraining of decision trees N Kumar, R Yalovetzky, C Li, P Minssen, M Pistoia Quantum 9, 1588	8	2025
Adaptive and robust watermark for generative tabular data DD Ngo, A Ray, A Seshadri, D Scott, S Obitayo, N Kumar, VK Potluru, ...	5	2024

TITLE	CITED BY	YEAR
arXiv preprint arXiv:2409.14700		
Blind quantum machine learning with quantum bipartite correlator C Li, B Li, O Amer, R Shaydulin, S Chakrabarti, G Wang, H Xu, H Tang, ... Physical Review Letters 133 (12), 120602	22	2024
Characterizing barren plateaus in quantum ansätze with the adjoint representation E Fontana, D Herman, S Chakrabarti, N Kumar, R Yalovetzky, J Heredge, ... Nature Communications 15 (1), 7171	123	2024
Characterizing barren plateaus in quantum ansätze with the adjoint representation E Fontana, D Herman, S Chakrabarti, N Kumar, R Yalovetzky, J Heredge, ... Nature Communications 15 (1), 7171		2024
Privacy-preserving quantum federated learning via gradient hiding C Li, N Kumar, Z Song, S Chakrabarti, M Pistoia Quantum Science and Technology 9 (3), 035028	37	2024
Challenges and opportunities of quantum optimization in finance Z He, S Chakrabarti, D Herman, N Kumar, C Li, P Minssen, P Niroula, ... Proceedings of the 61st ACM/IEEE Design Automation Conference, 1-4	3	2024
Qc-forest: a classical-quantum algorithm to provably speedup retraining of random forest R Yalovetzky, N Kumar, C Li, M Pistoia arXiv preprint arXiv:2406.12008	2	2024
Evidence of scaling advantage for the quantum approximate optimization algorithm on a classically intractable problem R Shaydulin, C Li, S Chakrabarti, M DeCross, D Herman, N Kumar, ... Science Advances 10 (22), eadm6761	180	2024
Prospects of privacy advantage in quantum machine learning J Heredge, N Kumar, D Herman, S Chakrabarti, R Yalovetzky, ... arXiv preprint arXiv:2405.08801	10	2024
Danylo Lykov, Pierre Minssen, Yue Sun, Yuri Alexeev, Joan M R Shaydulin, C Li, S Chakrabarti, M DeCross, D Herman, N Kumar, ... Dreiling, John P. Gaebler, Thomas M. Gatterman, Justin A. Gerber, Kevin ...	15	2024
Computing integral transforms based on physics informed neural networks E Philip, N Kumar, VE ELFVING WO Patent WO2024003092A1		2024
Des-q: a quantum algorithm to construct and efficiently retrain decision trees for regression and binary classification R Yalovetzky, N Kumar, C Li, P Minssen, M Pistoia Bulletin of the American Physical Society		2024
Communincation-efficient blind quantum machine learning with quantum bipartite correlator C Li, B Li, O Amer, R Shayludin, S Chakrabarti, G Wang, H Xu, H Tang, ...		2024

TITLE	CITED BY	YEAR
APS March Meeting Abstracts 2024, J00. 286		
Expressive quantum circuits provide inherent privacy in federated learning N Kumar, J Heredge, C Li, SH Sureshababu, S Eloul, M Pistoia APS March Meeting Abstracts 2024, W51. 004	1	2024
Quantum option pricing via the Karhunen-Lo{\e}ve expansion A Prakash, Y Sun, S Chakrabarti, C Che, A Dandapani, D Herman, ... arXiv preprint arXiv:2402.10132	5	2024
Hardness of the maximum-independent-set problem on unit-disk graphs and prospects for quantum speedups RS Andrist, MJA Schuetz, P Minssen, R Yalovetzky, S Chakrabarti, ... Physical review research 5 (4), 043277	51	2023
Expressive variational quantum circuits provide inherent privacy in federated learning N Kumar, J Heredge, C Li, S Eloul, SH Sureshababu, M Pistoia arXiv preprint arXiv:2309.13002	14	2023
\$ Des \$- \$ q \$: a quantum algorithm to construct and efficiently retrain decision trees for regression and binary classification N Kumar, R Yalovetzky, C Li, P Minnsen, M Pistoia arXiv preprint arXiv:2309.09976	2	2023
The adjoint is all you need: Characterizing barren plateaus in quantum ans\''atze E Fontana, D Herman, S Chakrabarti, N Kumar, R Yalovetzky, J Heredge, ... arXiv preprint arXiv:2309.07902	75	2023
Variational quantum algorithm for unconstrained black box binary optimization: Application to feature selection C Zoufal, RV Mishmash, N Sharma, N Kumar, A Sheshadri, A Deshmukh, ... Quantum 7, 909	41	2023
Danylo Lykov, Pierre Minssen, Yue Sun, et al. Evidence of scaling advantage for the quantum approximate optimization algorithm on a classically intractable problem R Shaydulin, C Li, S Chakrabarti, M DeCross, D Herman, N Kumar, ... Science Advances 10, 22	19	2023
Graph neural network initialisation of quantum approximate optimisation N Jain, B Coyle, E Kashefi, N Kumar Quantum 6, 861	75	2022
On the connection between quantum pseudorandomness and quantum hardware assumptions M Doosti, N Kumar, E Kashefi, K Chakraborty Quantum Science & Technology 7 (3), 035004	7	2022
Integral transforms in a physics-informed (quantum) neural network setting: Applications & use-cases N Kumar, E Philip, VE Elfving	8	2022

TITLE	CITED BY	YEAR
arXiv preprint arXiv:2206.14184		
Progress toward practical quantum cryptanalysis by variational quantum cloning B Coyle, M Doosti, E Kashefi, N Kumar Physical Review A 105 (4), 042604	18	2022
Optimal quantum-programmable projective measurements with coherent states N Kumar, U Chabaud, E Kashefi, D Markham, E Diamanti Physical Review Research 3 (4), 043035	7	2021
Client-server identification protocols with quantum puf M Doosti, N Kumar, M Delavar, E Kashefi ACM Transactions on Quantum Computing 2 (3), 1-40	35	2021
Experimental demonstration of quantum advantage for NP verification F Centrone, N Kumar, E Diamanti, I Kerenidis CLEO: QELS_Fundamental Science, FTh2N. 5	2	2021
Quantum versus classical generative modelling in finance B Coyle, M Henderson, J Chan Jin Le, N Kumar, M Painsi, E Kashefi Quantum Science & Technology 6 (2), 024013	87	2021
Experimental demonstration of quantum advantage for NP verification with limited information F Centrone, N Kumar, E Diamanti, I Kerenidis Nature communications 12 (1), 850	31	2021
Efficient construction of quantum physical unclonable functions with unitary t-designs N Kumar, R Mezher, E Kashefi arXiv preprint arXiv:2101.05692	18	2021
Variational quantum cloning: Improving practicality for quantum cryptanalysis B Coyle, M Doosti, E Kashefi, N Kumar arXiv preprint arXiv:2012.11424	10	2020
Practically feasible robust quantum money with classical verification N Kumar Cryptography 3 (4), 26	21	2019
Experimental demonstration of quantum advantage for one-way communication complexity surpassing best-known classical protocol N Kumar, I Kerenidis, E Diamanti Nature communications 10 (1), 4152	36	2019
Quantum communication advantage with coherent states and one beam splitter F Centrone, N Kumar, E Diamanti, I Kerenidis Quantum Information and Measurement, F5A. 40		2019

TITLE	CITED BY	YEAR
Efficient quantum communications with coherent state fingerprints over multiple channels N Kumar, E Diamanti, I Kerenidis Physical Review A 95 (3), 032337	40	2017
Nonlocality and conflicting interest games A Pappa, N Kumar, T Lawson, M Santha, S Zhang, E Diamanti, I Kerenidis Physical review letters 114 (2), 020401	96	2015
Quantum fingerprinting with coherent states for multiple clients N Kumar, E Diamanti, I Kerenidis GdR" Information Quantique: Fondements et Applications" Colloquium		2015
Quantum algorithm to solve a maze: converting the maze problem into a search problem N Kumar, D Goswami arXiv preprint arXiv:1312.4116	8	2013
Experimental demonstration of quantum advantage for one-way communication complexity N Kumar, I Kerenidis, E Diamanti arXiv preprint arXiv:1811.09154	3	1811
& Pistoia, M.(2024). Evidence of scaling advantage for the quantum approximate optimization algorithm on a classically intractable problem R Shaydulin, C Li, S Chakrabarti, M DeCross, D Herman, N Kumar Science Advances 10, 22	40	
Blind quantum machine learning with quantum bipartite correlator (2023) C Li, B Li, O Amer, R Shaydulin, S Chakrabarti, G Wang, H Xu, H Tang, ... arXiv preprint arXiv:2310.12893	2	
The adjoint is all you need: Characterizing barren plateaus in quantum ansätze (2023) E Fontana, D Herman, S Chakrabarti, N Kumar, R Yalovetzky, J Heredge, ... arXiv preprint arXiv:2309.07902	13	